Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Mountain View, CA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with backend distributed systems experience building scalable, low-latency services and APIs in Java, Python, C/C++, and Go across AWS and Kubernetes. Skilled in high-reliability infrastructure, cross-functional delivery, and production debugging.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript, TypeScript, SQL, Go.

Backend & Distributed Systems: REST APIs, Microservices, Node.js, Django, Flask, FastAPI, Low-Latency Service Design.

Databases: PostgreSQL, MySQL, Redis.

Infrastructure & Reliability: AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, GitLab CI/CD, Linux, Bash, SonarQube.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage distributed automation pipeline decoupling video understanding from device control for Android bug reproduction, evaluated against **ViBR** baseline across 53 runs on 20+ apps.
- Cut end-to-end latency **25.7x** (39.4s vs 1012.1s avg per run) by removing redundant per-scene inference, preserving task-completion fidelity under real-time constraints.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Built backend distributed automation services in Python against AWS (EC2, S3), replacing legacy scripts and cutting deployment time **40%** across production environments.
- Owned **44 production upgrades** in 4 months, diagnosing live infrastructure faults across Linux, EC2, and databases to keep user-facing services highly available.
- Designed **Ansible-based Blue-Green deployment** framework ensuring fault-tolerant rollout with reliable rollback, cutting downtime **50%**.
- Migrated **40+ production environments** to Kubernetes with ArgoCD and Helm, architecting scalable service infrastructure for internal and partner-facing systems.
- Served as Quality Coach, raising automated code coverage **35% to 78%**; mentored **5 engineers** while collaborating cross-functionally with product and QA.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built distributed ML system forecasting ride demand across **263 zones** (**2.3M-row** dataset) via low-latency **REST API**; two-stage model cut RMSE **43%** over baseline.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, MCP] May 2026

- Architected **17-tool MCP server** with Zod-validated APIs exposing structured tool-calling for external agents, backed by evidence-grounded RAG pipeline over Google Gemini.
- Built full-stack TypeScript backend (Express REST API + MCP server) unifying multi-payer eligibility workflows on a single scalable Node.js service.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).